

Gemma 3 and Gemma 4 results: circle topology at temperature 0

Source files

Model	Study report	Detailed circle/temperature-0 report
Gemma 3	Clean 50-trial data	Circle, temperature 0
Gemma 4 (gemma4:e2b-it-qat)	Leavitt study	Circle, temperature 0

Experimental settings

Setting	Gemma 3	Gemma 4
Topology	Circle	Circle
Temperature	0	0
Number of trials	50	50

Outcome results

Metric	Gemma 3	Gemma 4
Successful trials	10/50	19/50
Success rate	20.0%	38.0%
Average total messages	12.36	17.44

Time comparison

Total trial time is measured from each trial's recorded start timestamp through its completion timestamp. Experiment time begins with the first valid experiment message. The total and average rows use the same 50 circle-topology trials at temperature 0.

Timing metric	Gemma 3	Gemma 4
Total time for 50 trials	3,147.45 s (52 min 27.45 s)	40,555.71 s (11 h 15 min 55.71 s)
Average total time per trial	62.95 s	811.11 s
Longest total trial	265.22 s (trial 1)	1,736.27 s (trial 26)
Total experiment time for 50 trials	2,867.64 s (47 min 47.64 s)	39,171.51 s (10 h 52 min 51.51 s)
Average experiment time per trial	57.35 s	783.43 s
Longest experiment time	214.90 s (trial 1)	1,663.18 s (trial 26)
Average time per recorded response	Not recorded	43.484 s (922 responses)
Longest recorded response	Not recorded	120.412 s (Agent5, trial 7)

Data availability note: Gemma 3 did not store individual response durations. Its message elapsed values omit the first response and include inter-turn scheduling time, so they are not used as a response-time estimate.

Recipient-choice results

Sender	Recipient	Gemma 3 choices	Gemma 3 probability	Gemma 4 choices	Gemma 4 probability
Agent1	Agent2	10/138	7.2%	95/188	50.5%
Agent1	Agent5	128/138	92.8%	93/188	49.5%
Agent2	Agent1	102/131	77.9%	87/172	50.6%
Agent2	Agent3	29/131	22.1%	85/172	49.4%
Agent3	Agent2	107/110	97.3%	78/166	47.0%
Agent3	Agent4	3/110	2.7%	88/166	53.0%
Agent4	Agent3	84/117	71.8%	104/180	57.8%
Agent4	Agent5	33/117	28.2%	76/180	42.2%
Agent5	Agent1	90/122	73.8%	76/164	46.3%
Agent5	Agent4	32/122	26.2%	88/164	53.7%

Fallback routes

Model	Sender	Fallback source	Count
Gemma 3	-	No fallback routes recorded	0
Gemma 4	Agent2	server_assigned	1
Gemma 4	Agent5	server_assigned	1